MIL-T-37413A
27 September 1977
SUPERSEDING
MIL-T-37413(DSA-DM)
15 March 1976

MILITARY SPECIFICATION

TRETINOIN CREAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Tretinoin Cream.

1.2 Classification. The cream covered by this specification shall be of the following strengths (see 6.2):

Strength 1 - 0.1 percent
Strength 2 - 0.05 percent

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a bland hydrophilic cream base containing Tretinoin, in the following amounts:

Strength 1 - Shall contain 0.1 percent tretinoin.
Strength 2 - Shall contain 0.05 percent tretinoin.

3.1.1 Assay. The finished cream shall contain not less than 90.0 percent and not more than 125.0 percent of the required amount of tretinoin calculated on a weight/weight basis when determined as specified in 4.8.1.

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3.1.2 Bacterial count. The finished cream shall contain not more than 50 micro-organisms per gram. None of the following organisms shall be present: Staphylococcus aureus, fecal Streptococci, Candida albicans, Pseudomonas species, members of the Enterobacteriaceae. Presence or absence of micro-organisms shall be determined by appropriate, accurate, reproducible method(s).

3.2 Ingredients.

3.2.1 Tretinoin. The tretinoin powder used in the manufacture of the cream shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the cream shall be of USP or NF quality, or if not included in either of these compendia they shall be suitable for use in the cream.

3.4 Expiration dating period. The expiration dating period shall be not less than 24 months.

3.5 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

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4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of cream produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105, with an inspection level of S-2 and an acceptable quality level (AQL) of 1.0 percent defective for major defects, and 2.5 percent defective for minor defects.

4.4 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE I. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay	3.1.1	S-2	5
Bacterial count	3.1.2	-	

1/ Where possible, the sample shall be used for two or more tests.

4.5 Examination. The finished cream shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.4.

4.5.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE II. Classification of defects.

Categories	Defects 1/
Major	
101	Cream not free from foreign matter.
102	Cream not uniform in color.
103	Cream not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

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4.6 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Labeling inspection. Sampling shall be in accordance with MIL-STD-105, and the unit of product shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.8 Tests of finished cream.

4.8.1 Assay.

(Caution: Since tretinoin is a skin irritant, use adequate eye protection and avoid skin contact. If any of the material comes in contact with the skin wash well with soap and water.)

Tretinoin decomposes in the presence of light. Perform the analysis without undue delay, while protecting samples and standards from light. Tretinoin standards should be stored in tightly sealed vials at a constant temperature of $-10^{\circ} \pm 3^{\circ}$ C.

Solvent mixture. Prepare a 1:1 mixture of reagent grade chloroform and spectrograde isopropanol.

Standard solution. Prepare a solution of tretinoin reference standard with a known, final concentration of approximately 3.75 µg per ml in amber glassware (or otherwise protected from light).

Procedure. Accurately weigh an amount of finished cream approximately equivalent to 375 µg of tretinoin powder (375 mg of 0.1 percent cream; 750 mg of 0.05 percent cream) into a 100 ml tared, amber volumetric flask. Dissolve in and dilute to volume with the solvent mixture

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previously described. Centrifuge a portion of solution in centrifuge tubes, capped, and wrapped with aluminum foil to exclude light. Determine the absorbances of the standard, and clear, supernatant sample solution at the wavelength of maximum absorbance (approximately 358 nm) in a 1-cm cell, with a suitable spectrophotometer versus the solvent mixture as the blank.

Calculation:

$$\frac{Au}{As} \times \text{concentration standard (mg per ml)} \times \frac{100}{\text{sample weight (mg)}} \times 100 = \text{percent Tretinoin in cream W/W}$$

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One box containing 24 tubes of strength 1 or strength 2, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container: Strength 1 and strength 2. The immediate container shall be a commercially available ointment tube considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.9 of PPP-C-186, except pads and/or liners shall not be required.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

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5.3.1 Level A. Twelve (12) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve (12) units of issue shall be packed as specified in 5.4.3.3 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit packages. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.
(c) the expiration date.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)". Use of the legend: "STORE BELOW 80° F" is permissible in lieu of "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

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5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)". Use of the legend: "STORE BELOW 80° F" is permissible in lieu of "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-01-044-9388	TRETINOIN CREAM, 0.1%, 20 Gram, 24s	Strength 1
6505-01-044-9389	TRETINOIN CREAM, 0.05%, 20 Gram, 24s	Strength 2

6.3 Changes from previous issue. Asterisks to indicate changes have not been used in this specification due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA - DM

Project No. 6505-1882

Review activity:
DoD - MB

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-37413 A TRETINOIN CREAM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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